

KYIV SCHOOL OF ECONOMICS

Department of Economics · EBD BE

PROBABILITY & STATISTICS MIDTERM EXAMINATION

Summer Term, 2026 | Instructor: Rozora I.V. | Time: 90 minutes

Full Name: _____ **Score:** _____/100
Programme: _____ **Grade:** _____
Group: _____ **Date:** May __, 2026

Part	Section	Format	Items	Points
I	A Multiple Choice	circle one	5 questions	10
	B True or False	T/F + correct	5 statements	10
	C Validity Check	show work	1 function	5
	D R Output	interpret	5 questions	15
<i>Part I subtotal</i>				40
II	Problem 1	show work	sub-parts	30
	Problem 2	show work	sub-parts	30
<i>Part II subtotal</i>				60
TOTAL				100

Instructions: (1) Show all work in Part II. (2) Standard Normal Table and Distributions Summary on last pages. (3) No electronic devices or notes. (4) Write legibly.

	A	B	C	D	P 1	P 2	Total
Max	10	10	5	15	30	30	100
Score							

PART I: THEORY**(40 points)****SECTION A** Multiple Choice

[10 pts]

Circle the **one** correct answer. 5 questions, **2 points each**. No partial credit.

1. The conditional probability $P(A | B)$ is defined as:

- (A) $P(A) \cdot P(B)$
- (B) $\frac{P(A \cap B)}{P(B)}$
- (C) $\frac{P(B | A)}{P(A)}$
- (D) $1 - P(A^c | B)$

2. $X \sim \text{Exponential}(\lambda = 0.5)$. What is $\text{Var}(X)$?

- (A) 0.5
- (B) 2
- (C) 4
- (D) 0.25

3. Let X be any random variable with mean $E[X] = \mu$ and constants $a, b \in \mathbb{R}$. Which formula is always correct?

- (A) $E[aX + b] = aE[X]$
- (B) $E[aX + b] = E[X] + b$
- (C) $E[aX + b] = a\mu + b$
- (D) $E[aX + b] = a^2\mu + b$

4. Events A and B are **independent**. Which of the following must be **TRUE**?

- (A) $P(A \cap B) = P(A) + P(B)$
- (B) $P(A \cup B) = 1$
- (C) $P(A | B) = P(A)$
- (D) $P(A \cap B) = 0$

5. Which statement about the correlation coefficient ρ is **TRUE**?

- (A) $\rho > 0$ implies X causes Y to increase.
- (B) $\rho = 0$ implies X and Y are independent.
- (C) $|\rho|$ measures the strength of the *linear* association between X and Y .
- (D) ρ can exceed 1 for non-normal distributions.

SECTION B True or False

[10 pts]

Mark **T** (True) or **F** (False). If **FALSE**, write the corrected statement. 5 statements, 2 points each.

#	Statement	T / F	Correction if False
1	If A and B are mutually exclusive with $P(A) > 0$ and $P(B) > 0$, then A and B cannot be independent.		
2	For any two random variables X and Y : $E[X + Y] = E[X] + E[Y]$, regardless of dependence.		
3	The Poisson distribution is appropriate for modelling the number of events in a fixed time interval when events occur at a constant rate and independently.		
4	If $P(A B) = P(A)$, then events A and B are dependent.		
5	For a continuous random variable X , the PDF value $f(x)$ can exceed 1 for some values of x .		

SECTION C Validity Check

[5 pts]

Verify **both** conditions: (i) non-negativity, and (ii) integral equals 1. **Show all work.** State your conclusion clearly.

C1. [5 pts] Is the following a valid **PDF**?

$$f(x) = 6x(1 - x), \quad 0 \leq x \leq 1, \quad f(x) = 0 \text{ otherwise.}$$

Valid PDF? _____

Work:

C2. [+2 bonus pts] Find $k > 0$ such that $f(x) = k e^{-3x}$, $x \geq 0$, $f(x) = 0$ for $x < 0$, is a valid **PDF**. Verify both conditions for your k .

SECTION D R Output Interpretation

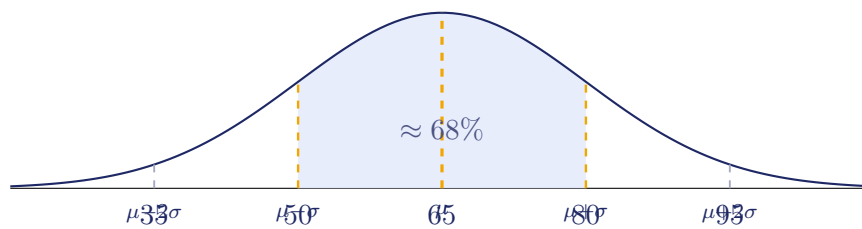
[15 pts]

Answer in complete sentences. **Interpretation only - no arithmetic required.** 5 questions, 3 points each.

After the midterm, a statistics professor models the score distribution of the entire KSE cohort as $X \sim N(\mu = 65, \sigma = 15)$ and runs the following analysis.

```
> # X = KSE midterm score, X ~ N(mu = 65, sd = 15)
> pnorm(80, mean = 65, sd = 15)
[1] 0.8413
> pnorm(50, mean = 65, sd = 15)
[1] 0.1587
> pnorm(95, mean = 65, sd = 15, lower.tail = FALSE)
[1] 0.0228
> qnorm(0.90, mean = 65, sd = 15)
[1] 84.22
> qnorm(0.10, mean = 65, sd = 15)
[1] 45.78
```

The plot below shows the distribution. The **shaded region** marks the interval $[\mu - \sigma; \mu + \sigma] = [50; 80]$. Use it to anchor your answers.



- (a) $\text{pnorm}(80, \dots) = 0.8413$ A student scored 80 on the midterm. What fraction of the cohort scored *below* them? Looking at the plot, does 80 appear to be a typical score or a high one? [3 pts]
- (b) $\text{pnorm}(50, \dots) = 0.1587$ The passing threshold is 50 points. What fraction of students failed? Locate 50 on the plot and express it in terms of μ and σ . [3 pts]
- (c) The shaded region on the plot represents $P(50 < X < 80) = 0.6826$. How is this calculated from the two **pnorm** outputs? Which well-known σ -rule does this value correspond to? [3 pts]

(d) `qnorm(0.90, ...)` = 84.22 The scholarship committee rewards the top 10% of students. What is the minimum score to qualify? On the plot, would this threshold fall to the *left* or *right* of $\mu + \sigma$? [3 pts]

(e) `qnorm(0.10, ...)` = 45.78 The tutoring centre contacts the bottom 10% of students. What is the score threshold? Using the plot, explain in one sentence why 45.78 is *below* the $\mu - \sigma$ line (50) rather than equal to it. [3 pts]

(f) **Bonus [+2 pts]** Using **only** the symmetry of the normal distribution and the outputs already shown - *no new table lookups or R calls* - determine $P(45.78 < X < 84.22)$. Show your reasoning in two or three sentences. (*Hint: what do the two quantile outputs together represent?*)

PART II: APPLIED PROBLEMS**(60 points)**

Show all work. Unjustified answers receive zero credit. Reference pages are at the end.

Problem 1 — The Pugs Scandal[30 pts]

The Kyiv International Dog Show uses a DNA test to verify that dogs entered in the *Pure Pug* category are genuine pugs (not pug-mix breeds). The breed registry provides the following data:

Known fact	Notation	Value
Share of registered “Pug” entries that are genuine pugs	$P(\text{Pug})$	0.20
Test correctly identifies a genuine pug (sensitivity)	$P(+ \mid \text{Pug})$	0.85
Test incorrectly flags a non-pug as positive (false positive rate)	$P(+ \mid \text{not Pug})$	0.10

(a) Using the **Law of Total Probability**, find $P(+)$: the overall probability that a randomly selected entry tests positive. [5 pts]

(b) Apply **Bayes’ theorem** to find $P(\text{Pug} \mid +)$: the probability that a dog with a positive test result is *actually* a genuine pug. [8 pts]

(c) Find $P(\text{not Pug} \mid -)$: the probability that a dog with a negative result is *truly* not a pug. Show the full application of Bayes’ theorem. [7 pts]

(d) The show organiser declares: “The test is 85% accurate, so a positive result means the dog is 85% likely to be a genuine pug.” Use your answer to (b) to evaluate this claim. Name the statistical fallacy and explain why it arises when the proportion of genuine pugs in the pool is small. [6 pts]

(e) A breeder submits **5 dogs** from the same kennel, tested independently. Each dog has probability $p = P(+)$ (from part a) of testing positive. Find the probability that **exactly 2** of the 5 dogs test positive. [4 pts]

(f) **Bonus [+3 pts] Two-stage protocol.** To reduce false positives, the show introduces a rule: a dog is disqualified only if it tests positive on *two independent* tests, each with the same characteristics. Using Bayes’ theorem, find $P(\text{Pug} \mid \text{both tests}+)$ and compare it with your answer from part (b). Does the two-stage protocol substantially improve diagnostic reliability? (Hint: treat “both tests positive” as your new event; recompute $P(\text{both}+)$ using the law of total probability.)

Problem 2 — Loto-Zabava[30 pts]

A Ukrainian state lottery sells scratch tickets at **10 UAH** each. Each ticket independently results in:

- **Win:** receive **90 UAH** (net gain +80 UAH) with probability $p = 0.10$.
- **Lose:** ticket is worthless (net gain -10 UAH) with probability 0.90 .

Let X_i denote the net gain from ticket i .

- (a) Compute $E[X_i]$ and $\text{Var}(X_i)$. Show all steps. State in one sentence whether buying a single ticket is a profitable decision on average. [6 pts]

- (b) A person buys $n = 100$ tickets over the course of a year. Let $S_{100} = \sum_{i=1}^{100} X_i$ be their total net gain.

- Compute $E[S_{100}]$ and $\text{Var}(S_{100})$.
- Name the theorem that justifies a normal approximation for S_{100} .
- State the approximate distribution of S_{100} with all numerical parameters.

[8 pts]

- (c) Use the CLT approximation to find $P(S_{100} > 0)$: the probability the player ends the year with a net profit. (Note: $\text{SD}(S_{100}) = 270$.) [8 pts]

(d) A devoted lottery player argues: “I’ve been buying tickets for years and I know that if I keep going long enough, I’ll eventually come out ahead overall.”

- Invoke the **Law of Large Numbers**: state precisely what $\bar{X}_n$ converges to as $n \rightarrow \infty$.
- Explain rigorously and in plain language why this makes the player’s optimism mathematically unfounded.

[8 pts]

(e) **Bonus [+3 pts]** Find the **minimum number of tickets** n such that $P(S_n > 0) \leq 0.25$, i.e. the player has at most a 25% chance of coming out ahead.

- Write $P(S_n > 0)$ in terms of n using the CLT (recall $E[X_i] = -1$, $\text{Var}(X_i) = 729$).
- Set up and solve the inequality.
- Interpret the result: after how many tickets does the player’s chance of profit drop to 25%?

(Use $z_{0.75} \approx 0.674$.)

REFERENCE PAGES

Distribution	PMF / PDF	Mean	Variance
Bernoulli(p)	$p^x(1-p)^{1-x}, x \in \{0, 1\}$	p	$p(1-p)$
Binomial(n, p)	$\binom{n}{k}p^k(1-p)^{n-k}$	np	$np(1-p)$
Geometric(p)	$(1-p)^{k-1}p, k = 1, 2, \dots$	$\frac{1}{p}$	$\frac{1-p}{p^2}$
Poisson(λ)	$\frac{\lambda^k e^{-\lambda}}{k!}, k \geq 0$	λ	λ
Uniform(a, b)	$\frac{1}{b-a}, a \leq x \leq b$	$\frac{a+b}{2}$	$\frac{(b-a)^2}{12}$
Exponential(λ)	$\lambda e^{-\lambda x}, x \geq 0$	$\frac{1}{\lambda}$	$\frac{1}{\lambda^2}$
Normal(μ, σ^2)	$\frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$	μ	σ^2

Standard Normal Cumulative Distribution: - $\Phi(z) = P(Z \leq z)$, $Z \sim N(0, 1)$

Area to the **left** of z . For negative z : $\Phi(-z) = 1 - \Phi(z)$. Example: $\Phi(-1.00) = 1 - 0.8413 = 0.1587$.

z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
0.0	.5000	.5040	.5080	.5120	.5160	.5199	.5239	.5279	.5319	.5359
0.1	.5398	.5438	.5478	.5517	.5557	.5596	.5636	.5675	.5714	.5753
0.2	.5793	.5832	.5871	.5910	.5948	.5987	.6026	.6064	.6103	.6141
0.3	.6179	.6217	.6255	.6293	.6331	.6368	.6406	.6443	.6480	.6517
0.4	.6554	.6591	.6628	.6664	.6700	.6736	.6772	.6808	.6844	.6879
0.5	.6915	.6950	.6985	.7019	.7054	.7088	.7123	.7157	.7190	.7224
0.6	.7257	.7291	.7324	.7357	.7389	.7422	.7454	.7486	.7517	.7549
0.7	.7580	.7611	.7642	.7673	.7704	.7734	.7764	.7794	.7823	.7852
0.8	.7881	.7910	.7939	.7967	.7995	.8023	.8051	.8078	.8106	.8133
0.9	.8159	.8186	.8212	.8238	.8264	.8289	.8315	.8340	.8365	.8389
1.0	.8413	.8438	.8461	.8485	.8508	.8531	.8554	.8577	.8599	.8621
1.1	.8643	.8665	.8686	.8708	.8729	.8749	.8770	.8790	.8810	.8830
1.2	.8849	.8869	.8888	.8907	.8925	.8944	.8962	.8980	.8997	.9015
1.3	.9032	.9049	.9066	.9082	.9099	.9115	.9131	.9147	.9162	.9177
1.4	.9192	.9207	.9222	.9236	.9251	.9265	.9279	.9292	.9306	.9319
1.5	.9332	.9345	.9357	.9370	.9382	.9394	.9406	.9418	.9429	.9441
1.6	.9452	.9463	.9474	.9484	.9495	.9505	.9515	.9525	.9535	.9545
1.7	.9554	.9564	.9573	.9582	.9591	.9599	.9608	.9616	.9625	.9633
1.8	.9641	.9649	.9656	.9664	.9671	.9678	.9686	.9693	.9699	.9706
1.9	.9713	.9719	.9726	.9732	.9738	.9744	.9750	.9756	.9761	.9767
2.0	.9772	.9778	.9783	.9788	.9793	.9798	.9803	.9808	.9812	.9817
2.1	.9821	.9826	.9830	.9834	.9838	.9842	.9846	.9850	.9854	.9857
2.2	.9861	.9864	.9868	.9871	.9875	.9878	.9881	.9884	.9887	.9890
2.3	.9893	.9896	.9898	.9901	.9904	.9906	.9909	.9911	.9913	.9916
2.4	.9918	.9920	.9922	.9925	.9927	.9929	.9931	.9932	.9934	.9936
2.5	.9938	.9940	.9941	.9943	.9945	.9946	.9948	.9949	.9951	.9952
2.6	.9953	.9955	.9956	.9957	.9959	.9960	.9961	.9962	.9963	.9964
2.7	.9965	.9966	.9967	.9968	.9969	.9970	.9971	.9972	.9973	.9974
2.8	.9974	.9975	.9976	.9977	.9977	.9978	.9979	.9979	.9980	.9981
2.9	.9981	.9982	.9982	.9983	.9984	.9984	.9985	.9985	.9986	.9986
3.0	.9987	.9987	.9987	.9988	.9988	.9989	.9989	.9989	.9990	.9990
3.4	.9997	.9997	.9997	.9997	.9997	.9997	.9997	.9997	.9997	.9998

Quantiles of the Standard Normal $z_p = \Phi^{-1}(p)$, i.e. $P(Z \leq z_p) = p$

p	0.90	0.95	0.975	0.99	0.995	0.999
z_p	1.282	1.645	1.960	2.326	2.576	3.090

Left tail: $\Phi^{-1}(1 - p) = -z_p$. *Example:* $\Phi^{-1}(0.05) = -1.645$.